

Methods Cheat Sheet

Jeremy Favro
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Revision 1

Power Series

For an ODE of the form $y'' + p(x)y' + q(x)y = 0$, the solution is of the form $\sum_{n=0}^{\infty} a_n x(x - x_0)^n$ if p, q are analytic at x_0 . This power series is guaranteed to converge for $|x - x_0| < R$ where R is the distance from x_0 to the nearest singular point.

Common Power Series

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ \cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \\ \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ \cosh x &= \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} \\ \sinh x &= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} \\ \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n \\ \frac{1}{1+x} &= \sum_{n=0}^{\infty} (-x)^n \\ \tan^{-1} x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \\ \ln(1+x)x &= \sum_{n=0}^{\infty} (-1)^{n+1} \frac{x^n}{n} \end{aligned}$$

Ordinary Points

A point is called ordinary if p and q are analytic at that point.

Solutions Around Singular Points

A point about which p, q are *not analytic* but $(x - x_0)p(x)$ and $(x - x_0)^2 q(x)$ are called regular singular. We are guaranteed one solution of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}$ where r is determined using the *indicial equation* $r(r-1) + [\lim_{x \rightarrow x_0} (x - x_0)p(x)]r + [\lim_{x \rightarrow x_0} (x - x_0)^2 q(x)] = 0$. We always use the larger r , r_1 first. This series is guaranteed to converge for

$0 < x - x_0 < R$. If the difference between the roots, $r_1 - r_2$ is not an integer then the series gives both linearly independent solutions. If the difference is an integer then the series gives at least one solution, with the other either give directly or given by $Cy_1 \ln(x) + \sum_{n=0}^{\infty} b_n (x - x_0)^{n+r_2}$. When the roots are repeated our second solution is guaranteed to be $y_1 \ln(x) + \sum_{n=0}^{\infty} b_n (x - x_0)^{n+r_2}$

Special Functions

Elementary Functions

Gamma Function

$$\begin{aligned} \Gamma(p) &= \int_0^{\infty} x^{p-1} e^{-x} dx \\ \Gamma(n+1) &= n! \forall n \in \mathbb{N} \\ \Gamma(p+1) &= p\Gamma(p) \\ \Gamma(1/2) &= \sqrt{\pi} \\ \Gamma(p)\Gamma(1-p) &= \frac{\pi}{\sin(\pi p)} \quad 0 < p < 1 \\ \Gamma(p)\Gamma(1-p) &= \int_0^{\infty} \frac{x^{p-1}}{1+x} dx \quad 0 < p < 1 \\ \Gamma(p) &\approx \sqrt{\frac{2\pi}{p}} \left(\frac{p}{e}\right)^p, p \rightarrow \infty \end{aligned}$$

Beta Function

$$\begin{aligned} \beta(p, q) &= \beta(q, p) = \int_0^1 x^{p-1} (1-x)^{q-1} dx, p, q > 0 \\ \beta(p, q) &= \frac{1}{(b-a)^{p+q-1}} \int_a^b (x-a)^{p-1} (b-x)^{q-1} dx \\ \beta(p, q) &= 2 \int_0^{\pi/2} \sin^{2p-1}(\theta) \cos^{2q-1}(\theta) d\theta \\ \beta(p, q) &= \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} \end{aligned}$$

Error Function

$$\begin{aligned} \operatorname{erf}(x) &= \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \\ \operatorname{erfc}(x) &= 1 - \operatorname{erf}(x) \\ \operatorname{erf}(0) &= 0 \\ \text{For } x \ll 1, \\ \operatorname{erf}(x) &\approx \frac{2}{\sqrt{\pi}} \left[x - \frac{x^3}{3} + \frac{x^5}{2!5} - \frac{x^7}{3!7} + \dots \right] \\ \text{For large } x, \\ \operatorname{erf}(x) &\approx \frac{e^{-x^2}}{x\sqrt{\pi}} \left[1 - \frac{1}{2x^2} + \frac{1 \cdot 3}{(2x^2)^2} + \dots \right]. \end{aligned}$$

Legendre Polynomials

ODEs of the form $(1-x^2)y'' - 2xy' + \ell(\ell+1)y = 0$ are

solved by $c_1 Q_{\ell}(x) + c_2 P_{\ell}(x)$.
 $P_{\ell}(x) = \frac{1}{2^{\ell}} \sum_{m=0}^{\ell} \frac{(-1)^m (2\ell-2m)!}{m!(\ell-m)!(\ell-2m)!}$

where $M = \begin{cases} \ell/2 & \ell \text{ even} \\ (\ell-1)/2 & \ell \text{ odd} \end{cases}$

$$P_{\ell}(x) = \frac{1}{2^{\ell} \ell!} \frac{d^{\ell}}{dx^{\ell}} (x^2 - 1)^{\ell}$$

Generating Function

Useful in proving recurrence relations

$$\Phi(x, h) = (1 - 2xh + h^2)^{-1/2} = \sum_{\ell=0}^{\infty} h^{\ell} P_{\ell}(x)$$

Recurrence Relations

$$\begin{aligned} (\ell+1)P_{\ell+1}(x) &= (2\ell+1)xP_{\ell}(x) - \ell P_{\ell-1}(x) \\ xP'_{\ell}(x) - P'_{\ell-1}(x) &= \ell P_{\ell}(x) \\ P'_{\ell+1}(x) - xP'_{\ell}(x) &= (\ell+1)P_{\ell}(x) \\ (2\ell+1)P_{\ell}(x) &= P'_{\ell+1}(x) - P'_{\ell-1}(x) \\ (1-x^2)P'_{\ell}(x) &= \ell P_{\ell-1}(x) - \ell x P_{\ell}(x) \end{aligned}$$

Orthogonality

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{2}{2n+1} \delta_{nm}$$

Bessel Functions

ODEs of the form $x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0$ are solved by $y = c_1 J_{\nu}(\lambda x) + c_2 Y_{\nu}(\lambda x)$.
 $J_{\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+\nu}}{2^{2m+\nu} \Gamma(m+\nu+1)}$
 $Y_{\nu}(x) = J_{\nu}(x) \ln(x) + \sum_{m=0}^{\infty} \frac{(-1)^m}{2^{2m} (m!)^2} x^{m+\nu} \cdot \sum_{n=1}^m \frac{1}{n}$.

ODEs of the form $x^2 y'' + xy' - (\lambda^2 x^2 + \nu^2)y = 0$ are solved by $y = c_1 I_{\nu}(\lambda x) + c_2 K_{\nu}(\lambda x)$.
 $I_{\nu}(x) = i^{-\nu} J_{\nu}(ix)$
 $K_{\nu}(x) = \frac{\pi}{2} \frac{I_{-\nu}(x) - I_{\nu}(x)}{\sin(\nu\pi)}$. I diverges for large x and K diverges for $x \rightarrow 0$. ODEs of the form

$x^2 y'' + (1-2a)xy' + (b^2 c^2 x^{2c} + (a^2 - p^2 c^2))y = 0$ are solved by $y = c_1 x^a J_p(bx^c) + c_2 x^a Y_p(bx^c)$.

Generating Function

Useful in proving recurrence relations

$$\Psi(x, h) = e^{\frac{1}{2}x(h-1/h)} = \sum_{n=-\infty}^{\infty} h^n J_n(x)$$

Properties

$$\begin{aligned} \frac{d}{dx} (x^{\nu} J_{\nu}(x)) &= x^{\nu} J_{\nu-1}(x) \\ \frac{d}{dx} (x^{-\nu} J_{\nu}(x)) &= -x^{-\nu} J_{\nu+1}(x) \\ J_{\nu-1}(x) + J_{\nu+1}(x) &= \frac{2\nu}{x} J_{\nu} \\ J'_{\nu} &= \frac{1}{2} [J_{\nu-1}(x) + J_{\nu+1}(x)] \\ xJ'_{\nu} &= \nu J_{\nu}(x) - xJ_{\nu+1} \\ J_{1/2}(x) &= \sqrt{\frac{2}{\pi x}} \sin(x) \\ J_{-1/2}(x) &= \sqrt{\frac{2}{\pi x}} \cos(x) \\ J_{3/2}(x) &= \sqrt{\frac{2}{\pi x}} \left[\frac{\sin(x)}{x} - \cos(x) \right] \\ J_{-3/2}(x) &= -\sqrt{\frac{2}{\pi x}} \left[\frac{\cos(x)}{x} + \sin(x) \right] \end{aligned}$$

Orthogonality

$\int_0^1 x J_{\nu}(a_n x) J_{\nu}(b_m x) dx = \frac{1}{2} (J'_{\nu}(a))^2 \delta_{nm}$ where a_n and b_m are the n th and m th zeroes of J_{ν} .

Self-Adjoint Operators and Sturm-Liouville

A Sturm-Liouville problem is given by $L[\phi] + \lambda w(x)\phi(x) = 0$. L is a self-adjoint linear differential operator. For $L[\phi] = P_0 \phi_{xx} + P_1 \phi_x + P_2 \phi$ this means that $P'_0 = P_1$. Distinct ϕ_n are orthogonal with respect to $w(x)$. The coefficients of the eigenfunction expansion for $f(x)$ are $a_n = \frac{\int_a^b w(x) f(x) \phi_n(x) dx}{\int_a^b w(x) \phi_n^2(x) dx}$. We can transform a non-self-adjoint L to a self adjoint one by multiplying by $P_0^{-1} \exp(\int P_1(x)/P_0(x) dx)$

Fourier Series

For some function $f(x)$ periodic with period $2L$:

$$\begin{aligned} a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\ a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\ \text{and} \\ f(x) &= a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \end{aligned}$$